

Engineering Physics Question bank with Solution

Chapter 1: Units and Measurement

Q 1. What is physical quantity?

Ans: A physical quantity is a quantity that can be measured and expressed using a number and a unit.

For example:

The length of a table is 2 metres.

The mass of a bag is 5 kilograms.

The time taken to run is 10 seconds.

The temperature of water is 30°C.

Q 2. Explain Fundamental physical quantity and Derived physical quantity.

Ans: 1. Fundamental Quantities

These are basic quantities that are independent of other physical quantities.

There are seven SI fundamental quantities:

Length — metre (m)

Mass — kilogram (kg)

Time — second (s)

Electric current — ampere (A)

Thermodynamic temperature — kelvin (K)

Amount of substance — mole (mol)

Luminous intensity — candela (cd)

2. Derived Quantities

These quantities are obtained by combining fundamental quantities mathematically.

Examples:

$$\text{Area} = \text{length} \times \text{length} \rightarrow \text{m}^2$$

$$\text{Volume} = \text{length} \times \text{length} \times \text{length} \rightarrow \text{m}^3$$

$$\text{Speed} = \text{distance} / \text{time} \rightarrow \text{m/s}$$

$$\text{Density} = \text{mass} / \text{volume} \rightarrow \text{kg/m}^3$$

$$\text{Force} = \text{mass} \times \text{acceleration} \rightarrow \text{newton (N)}$$

Q 3. Explain Scalar quantity and Vector quantity.

Ans: 1. Scalar Quantity

A scalar quantity is a physical quantity that has magnitude only and does not have any direction.

In simple words, if a quantity can be completely described by giving only a numerical value and a unit, it is called a scalar quantity.

Examples:

$$\text{Mass} = 5 \text{ kg}$$

$$\text{Time} = 10 \text{ s}$$

$$\text{Temperature} = 30^\circ\text{C}$$

$$\text{Distance} = 20 \text{ m}$$

$$\text{Speed} = 60 \text{ km/h}$$

$$\text{Volume} = 2 \text{ m}^3$$

$$\text{Energy} = 100 \text{ J}$$

For example, if we say "the mass of a box is 10 kg," the information is complete. We do not need to specify a direction. Therefore, mass is a scalar quantity.

2. Vector Quantity

A vector quantity is a physical quantity that has both magnitude and direction.

In simple words, a vector quantity requires a numerical value, a unit, and a direction to be completely described.

Examples:

Displacement

Velocity

Acceleration

Force

Momentum

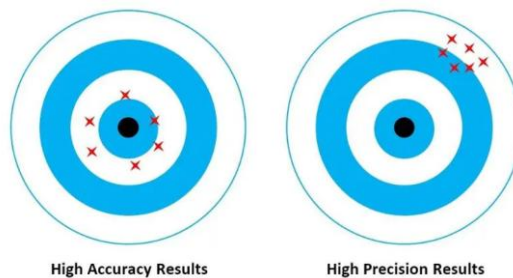
Q 4. Define Accuracy and Precision.

Ans: 1. Accuracy

Accuracy refers to how close a measured value is to the true or accepted value of a physical quantity.

In simple words:

Accuracy is the degree of closeness of a measured value to the true value.



Example:

Suppose the actual length of a rod is 10.00 cm.

If we measure it and obtain:

9.99 cm → very accurate

10.01 cm → very accurate

9.50 cm → less accurate

The closer the measured value is to 10.00 cm, the greater its accuracy.

Example from daily life:

If the actual temperature of a room is 25°C and a thermometer shows 25.1°C, the reading is highly accurate because it is very close to the actual temperature.

2. Precision

Precision refers to how close repeated measurements are to one another.

In simple words:

Precision is the degree of agreement or closeness among repeated measurements.

Example:

Suppose we measure the length of an object five times and obtain:

12.01 cm, 12.02 cm, 12.01 cm, 12.02 cm, 12.01 cm

These values are very close to each other. Therefore, the measurements are highly precise.

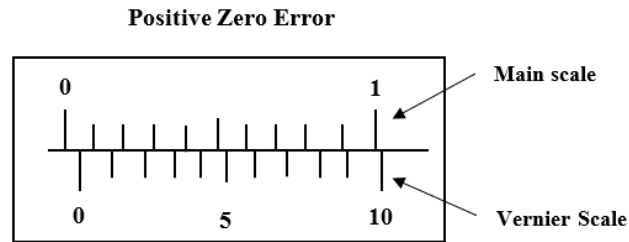
However, if the true length is 12.50 cm, these measurements are not accurate even though they are precise.

Q 5. What is positive error and Negative error in equipment?

Ans: 1. Positive Error

A positive error occurs when a measuring instrument gives a reading that is greater than the true or actual value.

Definition: Positive error is the error in which the measured value is greater than the true value.



It can be expressed as:

$$\text{Positive Error} = \text{Measured Value} - \text{True Value}$$

when the measured value is greater than the true value.

Example

Suppose the actual length of an object is 10.0 cm, but a measuring instrument shows 10.2 cm.

Therefore:

$$\text{Error} = 10.2 - 10.0 = +0.2 \text{ cm}$$

So, the instrument has a positive error of 0.2 cm.

Example in an Instrument

Consider a weighing balance that should show 0 kg when nothing is placed on it, but it shows +0.5 kg.

This means the balance has a positive zero error.

To obtain the correct reading, this error must be taken into account.

2. Negative Error

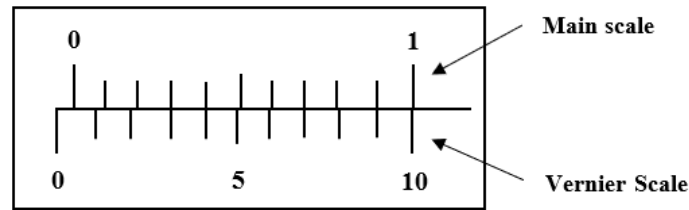
A negative error occurs when a measuring instrument gives a reading that is less than the true or actual value.

Definition: Negative error is the error in which the measured value is less than the true value.

It can be expressed as: $\text{Negative Error} = \text{Measured Value} - \text{True Value}$

when the measured value is less than the true value.

Negative Zero Error



Example

Suppose the actual length of an object is 10.0 cm, but the instrument shows 9.8 cm.

Therefore: Error = $9.8 - 10.0 = -0.2$ cm

So, the instrument has a negative error of 0.2 cm.

Example in an Instrument

Suppose a weighing balance should show 0 kg when it is empty, but it shows -0.5 kg.

This means the balance has a negative zero error.

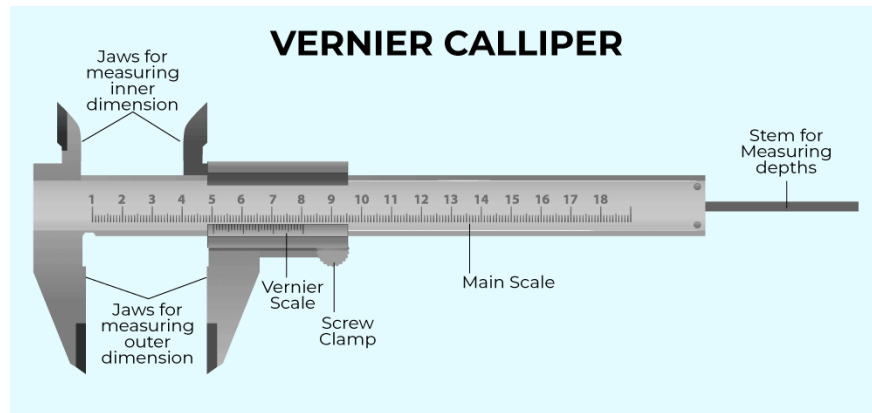
Q 6. Explain vernier cliper with sketch.

Ans: A Vernier caliper is a precision measuring instrument used to measure the length, diameter, thickness, and depth of an object accurately. It can measure both external and internal dimensions.

It is commonly used in physics laboratories, mechanical workshops, and engineering work.

Main Parts of a Vernier Caliper

A vernier caliper consists of the following main parts:



1. Main Scale

The main scale is a graduated scale marked in centimetres and millimetres. It is fixed and provides the main measurement.

2. Vernier Scale

The vernier scale is a small movable scale that slides along the main scale. It helps us measure fractions of the smallest main-scale division.

3. Outside Jaws

The large jaws at the bottom are called outside jaws. They are used to measure:

External diameter of a cylinder

External diameter of a pipe

Thickness of an object

Length or width of an object

4. Inside Jaws

The small jaws at the top are called inside jaws. They are used to measure the internal diameter of objects such as pipes, holes, and hollow cylinders.

5. Depth Rod

A thin rod extending from the end of the caliper is called the depth rod. It is used to measure the depth of holes, containers, and narrow spaces.

6. Locking Screw

The locking screw is used to hold the sliding vernier scale in position after taking a measurement.

7. Fine Adjustment Screw

The fine adjustment mechanism allows the movable jaw to be adjusted slowly and accurately.

Principle of Vernier Caliper

The vernier caliper works on the Vernier principle.

The basic idea is that the divisions on the vernier scale are slightly different from the divisions on the main scale. By observing which vernier division coincides with a main-scale division, we can measure a small fraction of the smallest main-scale division.

Least Count

The least count of a vernier caliper is the smallest measurement that can be accurately measured by it.

A common formula is:

Least Count = 1 Main Scale Division – 1 Vernier Scale Division

For example, if: 1 MSD = 1 mm and 10 VSD = 9 mm

then: 1 VSD = 0.9 mm

Therefore: Least Count = $1.0 - 0.9 = 0.1$ mm

So the least count is: 0.1 mm = 0.01 cm

How to Measure Using a Vernier Caliper

The general procedure is:

Step 1: Check Zero Error

Close the jaws gently and check whether the zero of the main scale and the zero of the vernier scale coincide.

If they do not coincide, there is a zero error, which must be considered.

Step 2: Place the Object

- Place the object between the appropriate jaws.
- Use outside jaws for external measurements.
- Use inside jaws for internal measurements.
- Use the depth rod for depth measurements.

Step 3: Adjust the Jaws

Move the sliding jaw until it just touches the object. Do not apply excessive pressure.

Step 4: Take the Main Scale Reading

Note the main-scale reading just to the left of the zero of the vernier scale.=

Step 5: Find the Vernier Coincidence

Look at the vernier scale and find the division that exactly coincides with a division on the main scale.

Step 6: Calculate the Reading

The observed reading is generally calculated as:

Reading = Main Scale Reading + (Vernier Coincidence $\times$ Least Count)

If zero correction is required:

Corrected Reading = Observed Reading + Zero Correction

Example

Suppose:

- Main Scale Reading = 2.3 cm
- Vernier coincidence = 6
- Least Count = 0.01 cm

Then: Vernier Reading = $6 \times 0.01 = 0.06$ cm

Therefore: Total Reading = $2.3 + 0.06$

= 2.36 cm

So, the measured value is 2.36 cm, before applying any zero correction.

Uses of Vernier Caliper

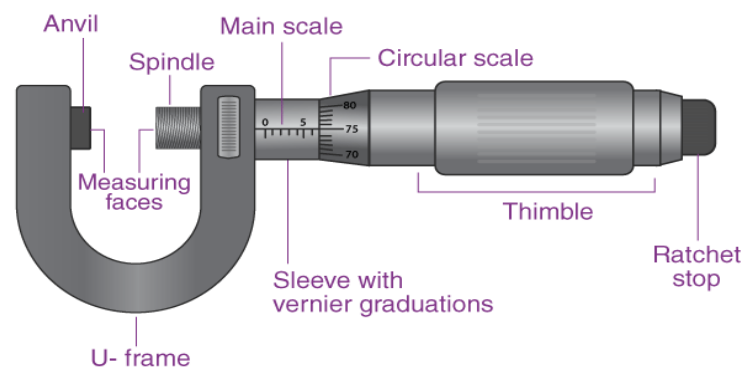
1. A vernier caliper can be used to measure:
2. External diameter of a cylinder.
3. Internal diameter of a hollow pipe.
4. Thickness of a sheet.
5. Length and width of small objects.
6. Depth of a hole or container.
7. Diameter of a small spherical or cylindrical object.

Q 7. Explain Micrometer Screw gauge.

Ans: Definition

A micrometer screw gauge is a precision instrument that works on the principle of a screw and is used to measure very small dimensions accurately.

Main Parts of a Micrometer Screw Gauge



1. Frame

The frame is the strong, U-shaped body of the screw gauge. It supports the other parts and provides a rigid structure for measurement.

2. Anvil

The anvil is a fixed, smooth measuring surface.

The object to be measured is placed between the anvil and spindle.

3. Spindle

The spindle is a movable measuring rod. It moves towards or away from the anvil when the thimble is rotated.

4. Sleeve or Main Scale

The sleeve contains the main scale, usually graduated in millimetres and sometimes half-millimetres.

It gives the main part of the measurement.

5. Thimble

The thimble is the rotating part of the instrument. It carries the circular or thimble scale.

When the thimble is rotated, the spindle moves.

6. Ratchet

The ratchet is used to apply a small and nearly constant force while measuring.

It prevents the object from being squeezed too tightly, which could cause an incorrect reading.

7. Locking Screw

The locking screw is used to hold the spindle in position after the object has been measured.

Principle of Micrometer Screw Gauge

The micrometer screw gauge works on the principle of the screw.

When a screw is rotated, it moves forward or backward along its axis by a small distance. This property is used to measure very small lengths accurately.

The distance moved by the spindle during one complete rotation of the screw is called the pitch.

Pitch

Pitch is the distance moved by the spindle in one complete rotation of the screw.

For example, if the spindle moves 0.5 mm in one complete rotation:

Pitch = 0.5 mm

How to Take a Measurement

The following steps are generally followed:

Step 1: Check Zero Error

First, bring the spindle into contact with the anvil gently using the ratchet.

Check whether the zero of the circular scale coincides with the reference line on the sleeve.

If they do not coincide, the instrument has a zero error.

Step 2: Place the Object

Place the object, such as a wire, between the anvil and spindle.

Step 3: Rotate the Thimble

Rotate the thimble until the spindle gently touches the object.

Use the ratchet to avoid applying excessive force.

Step 4: Note the Main Scale Reading

Observe the reading on the sleeve just before the edge of the thimble.

This is the main scale reading.

Step 5: Note the Circular Scale Reading

Find the circular-scale division that coincides with the reference line on the sleeve.

This is the circular scale reading.

Step 6: Calculate the Measurement

The observed reading can be calculated as:

Observed Reading = Main Scale Reading + (Circular Scale Reading $\times$ Least Count)

If there is zero error, apply the appropriate zero correction.